Who Is Struggling Financially in Kenya and Why?

A multiclass ML approach using the 2024 FinAccess Household Survey (20,871 adults across all 47 counties).

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52.6%

of Kenyans say finances worsened

9.9%

fully excluded from financial services

20,871

adults surveyed across 47 counties

Dataset & Exploratory Data Analysis

28 Features

Demographics, income, savings, loans, mobile access, financial literacy & shock exposure

5 Duplicates Removed

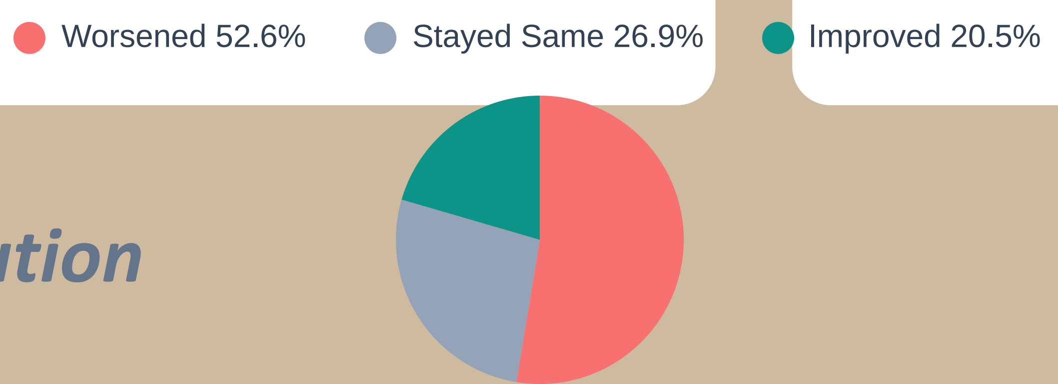
Cleaned to 20,863 records. Only 1 column had nulls: barriers_bank (27.5% → filled as 'Not Applicable')

Imbalanced Target

Worsened 52.6% · Stayed Same 26.9% · Improved 20.5% demanded a weighted F1 metric

Right-Skewed Numerics

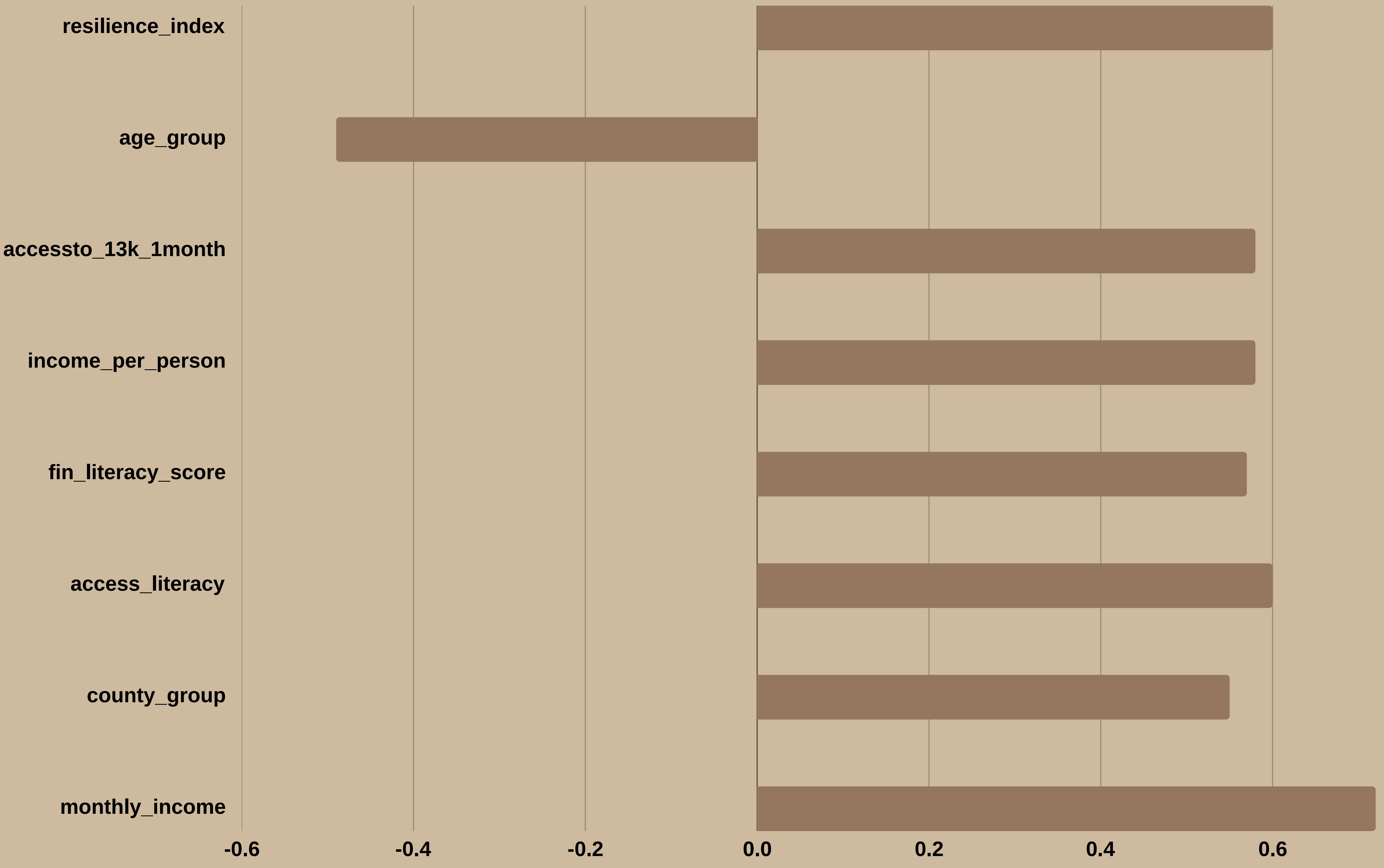
monthly_income, household_size, prodsum1 clipped at 95th pct & Yeo-Johnson transformed



Target Class Distribution

Key EDA Finding

'Stayed the Same' overlaps both extremes which is the core challenge of this dataset



Income drives outcomes

monthly_income has the strongest correlation (r = 0.72). Income instability is the #1 predictor.

Age is a vulnerability marker

Older adults (55+) show the strongest negative correlation (r = −0.49) .It’s most at risk.

'Stayed the Same' = noise

LDA & KMeans showed this class overlaps both Improved and Worsened, causing model confusion.

Modelling Approach

A principled pipeline to handle class overlap and extract clean signals

01

Feature Engineering

financial_literacy_score, resilience_index, access_literacy, income_per_person, shock_income are among the 16 final features

02

Baseline Models

Logistic Regression & Random Forest on full 3-class target → Accuracy ≈ 48–52%, F1 macro ≈ 0.45

03

LDA Separation

Linear Discriminant Analysis revealed strict Stable (LDA < −2) and strict Not Stable (LDA > 1) zones

04

Noise Removal

Extracted 3,235 clean-signal rows + 3,885 overlap ('Stayed Same') rows → 6,917 training records

05

Final Model

Logistic Regression on clean dataset with stratified 70/30 split → Weighted F1 = 0.98, Accuracy = 98%

06

Explainability

SHAP values computed per class : access_literacy, resilience_index, income dominate all three classes

Model Performance

Logistic Regression on noise-filtered dataset : Weighted F1 = 0.98

0.98
Weighted F1-Score

98%
Test Accuracy

6,917
Clean training samples

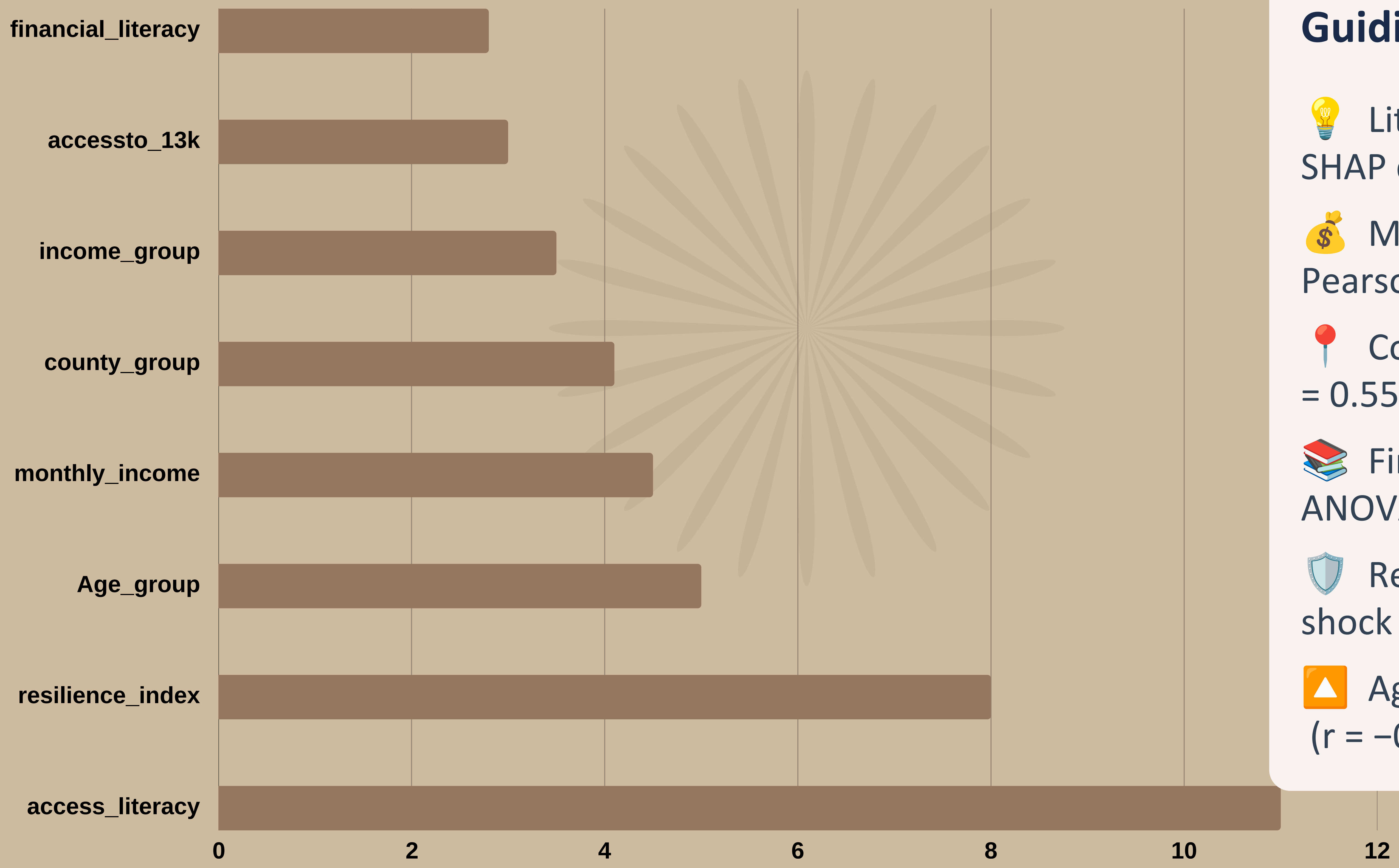
Class	Precision	Recall	F1-Score	Support
Worsened (0)	0.98	0.99	0.99	1,120
Stayed Same (1)	0.98	0.96	0.97	859
Improved (2)	0.94	0.94	0.94	97
Weighted Avg	**0.98**	**0.98**	**0.98**	**2,076**

Confusion Matrix Highlights

- ✔ 1,113 / 1,120 Worsened correct
- ✔ 828 / 859 Stayed Same correc
- ✔ 91 / 97 Improved correct
- Only 31 misclassifications total ROC-AUC ≈ 0.71 on full dataset

Key Drivers of Financial Outcomes

SHAP global importance · Correlation · ANOVA (p ≈ 0.0 for income & literacy)



Guiding Question Answer

💡 Literacy access (access_literacy) : top SHAP driver across ALL classes

💰 Monthly income : ANOVA p = 0.0; Pearson r = 0.72

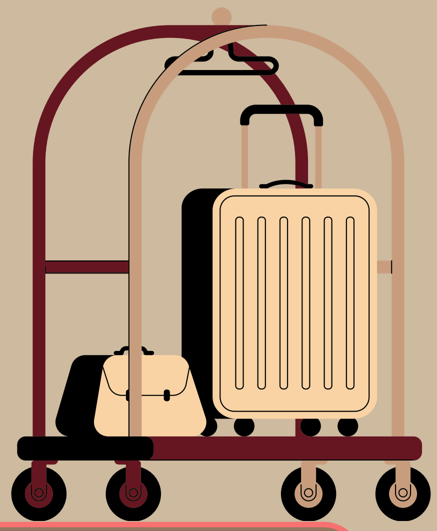
📍 County group : geographic disparity; r = 0.55

📚 Financial literacy score : r = 0.57; ANOVA p = 0.0

🛡️ Resilience index : literacy + inclusion + shock readiness

🔼 Age group : older adults most at risk (r = −0.49)

Recommendations



Boosting literacy, resilience, and regional equity will have the greatest impact on financial wellbeing

Policymakers

- Expand financial literacy programs in vulnerable counties and for adults 45+
- Invest in shock-preparedness safety nets (insurance, emergency funds)
- Reduce county-level disparities through regional equity investment

Banks & Fintechs

- Design inclusive savings & lending products for low-income households
- Use resilience-based scoring (literacy + inclusion + shock buffers) for credit
- Partner with mobile money providers to expand affordable access channels

NGOs & Development Partners

- Prioritise community-level literacy and resilience training
- Support income diversification programs to reduce shock vulnerability
- Target women and households with disabilities for tailored interventions

Financial deterioration is most strongly predicted by low literacy access, weak resilience, income instability, and regional disparities.